

Applied Statistics (MATH10096)

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r^2 : How close is the fit?

- ↪ r^2 measures the proportion (or percentage) of the variance in the original data that is 'explained' by the fitted model.
- ↪ The proportion of variance left unexplained, is the observed variance of the residuals, divided by the observed variance of the response data.
- ↪ The proportion of variance explained is hence one minus the same ratio, so

$$r^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2 / n}{\sum_i (y_i - \bar{y})^2 / n} = 1 - \frac{\sum_i (y_i - \hat{\mu}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

- ↪ The denominator can be thought as the observed variance of the residuals of a regression model that contains only an intercept term.
- ↪ Because the residual sum of squares, when a predictor is added, can not increase, and the denominator is the same for different models fitted to the same dataset, we conclude that when predictors are added to a model, r^2 is monotone increasing.

r^2 : How close is the fit?

- ↪ For exactly this reason, we have that $\sum_i \hat{\epsilon}_i^2 \leq \sum_i (y_i - \bar{y})^2$ thus implying that $(\sum_i \hat{\epsilon}_i^2 / n) / (\sum_i (y_i - \bar{y})^2 / n) \leq 1$ and therefore, $0 \leq r^2 \leq 1$.
- ↪ Further notice that this conventional definition uses biased variance estimators and, as a result, r^2 tends to overestimate how well a model is doing.
- ↪ The *adjusted* r^2 avoids this, to some extent, by using unbiased estimators:

$$r_{\text{adj}}^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2 / (n - p)}{\sum_i (y_i - \bar{y})^2 / (n - 1)}.$$

- ↪ r_{adj}^2 does not necessarily increase monotonically as predictors are added to a model.
- ↪ Note that r_{adj}^2 can be negative, which occurs when an estimated model predicts worse than an intercept only model.

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- ↪ A high r_{adj}^2 is always better than a low r_{adj}^2 (provided that the model assumptions are met).
- ↪ However, a low r_{adj}^2 should not necessarily be taken as indicating that a model is poor: some response data simply contain a good deal of random variability about which nothing can be done.
- ↪ Note that r^2/r_{adj}^2 does not measure the correctness of a linear model but its usefulness, assuming the model is correct.
- ↪ Note that when a model does not contain an intercept, R uses 0 in place of $\bar{y}$ and thus r^2 (artificially) increases substantially.

r^2 : How close is the fit?

```
library(MASS)
m4 <- lm(loss ~ tens + I(hard*tens) + I(tens^2) +
          I(hard^2*tens) + I(tens^3) + I(hard^3),
          data = Rubber)
summary(m4)

##
## Call:
## lm(formula = loss ~ tens + I(hard * tens) + I(tens^2) + I(hard^2 *
##    tens) + I(tens^3) + I(hard^3), data = Rubber)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.690  -7.647   2.943  12.765  50.057
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.252e+03  1.090e+03  -1.148  0.262582
## tens         3.810e+01  1.786e+01   2.133  0.043794 *
## I(hard * tens) -8.193e-02  3.443e-02  -2.379  0.026016 *
## I(tens^2)      -2.319e-01  9.656e-02  -2.402  0.024789 *
## I(hard^2 * tens) 8.105e-04  3.090e-04   2.623  0.015201 *
## I(tens^3)       4.574e-04  1.750e-04   2.614  0.015507 *
## I(hard^3)      -8.090e-04  2.072e-04  -3.905  0.000713 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.67 on 23 degrees of freedom
## Multiple R-squared:  0.9217, Adjusted R-squared:  0.9013
## F-statistic: 45.14 on 6 and 23 DF,  p-value: 1.375e-11
```